

Perceptual Thresholds of Sample Counts in AI-Augmented 3D Gaussian Splatting

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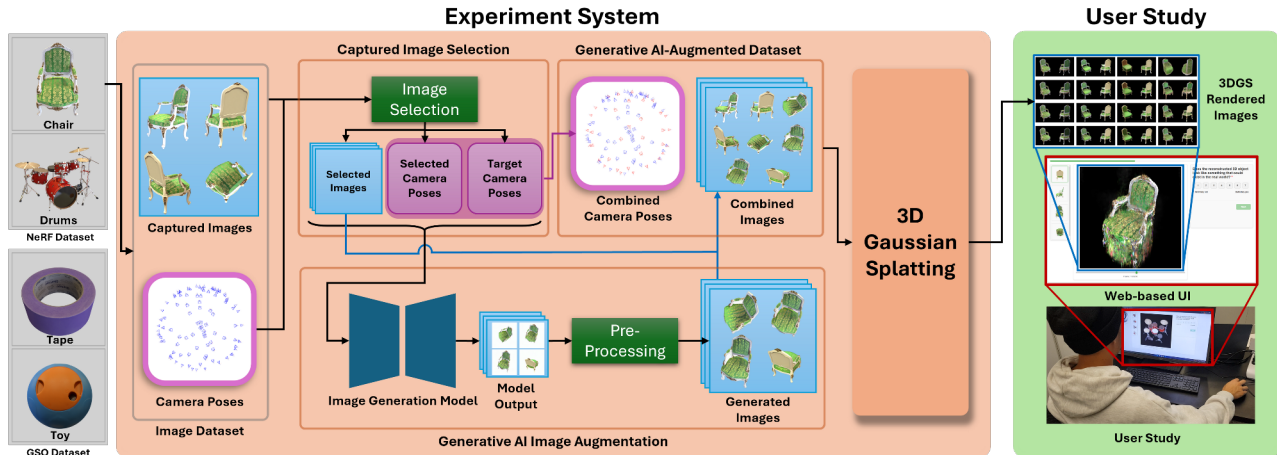


Figure 1: Overview of our experimental setup designed to investigate how input image sample count influences the perceived quality of image-based 3D reconstructions augmented by an image generation model.

ABSTRACT

Image-based 3D reconstruction methods such as 3D Gaussian Splatting (3DGS) are increasingly used to enable real-time free-viewpoint navigation in VR/AR, but achieving accurate texture and geometry typically requires dense multi-view captures. Recent generative models can synthesize additional training views from sparse inputs, yet it remains unclear how many generated views are needed and when augmentation helps or even degrades for 3DGS training for the perceived quality of 3DGS rendering. We conduct a perceptual user study that varies the number of captured and AI-generated images for 3DGS training, and identify perceptual thresholds: adding generated views improves plausibility and visual quality when real images are scarce, but provides diminishing or negative returns once real-view coverage becomes sufficient.

Index Terms: 3D Gaussian Splatting, Generative Image Augmentation, Visual Perception

1 INTRODUCTION

Recent image-based reconstruction and neural rendering offer an appealing solution by turning casual multi-view capture into a navigable 3D representation. Among recent approaches, 3D Gaussian Splatting (3DGS) [4] has emerged as a practical choice due to its efficient optimization and real-time rendering, which aligns well with interactive viewing in VR and AR.

A persistent challenge, however, is the reliance on dense and well-covered image observations. When only a small set of images is available, limited viewpoint coverage often leads to missing details and unstable appearance and perceptual artifacts during novel view rendering. [3] While collecting additional samples can mitigate these issues, capture cost and accessibility remain key constraints. To reduce capture demands, generative view synthesis has been recently explored as a data-centric strategy that expands training sets by generating novel views conditioned on existing images [2]. Despite rapid progress in reconstruction and generation, it remains unclear *when* and *how much* generative augmentation improves perceived 3DGS quality.

In this work, we present a perceptual user study that systematically examines the effect of generative view augmentation on 3DGS under controlled few-shot capture conditions. By varying the number of captured and AI-generated training images, we identify clear perceptual thresholds: augmentation improves perceived plausibility and visual quality when real images are scarce, but offers diminishing or negative returns once real-view coverage becomes sufficient. These findings provide actionable guidance for XR practitioners on when to apply generative augmentation and when to prioritize additional real captures.

2 EXPERIMENTAL DESIGN AND SETUP

We hypothesize that when the number of captured training images is limited, augmenting the dataset by generating additional views conditioned on the existing images can improve the perceptual quality of 3DGS reconstructions. To test this hypothesis under diverse texture and geometric complexity, we selected four objects, two from the NeRF synthetic dataset [6] and two from the Google Scanned Objects (GSO) dataset [1], as shown in Figure 1.

Based on these datasets, we constructed sparse training sets using greedy farthest-point sampling over camera viewpoints to maximize coverage (Figure 1).

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To generate missing views from these sparse image datasets, we used a generative model called EscherNet [5]. This model takes as input the reference images and camera poses, along with the target camera poses, and generates images rendered from those target viewpoints. We selected the target camera poses from the set of unsampled poses in the original dataset, again using a greedy algorithm. With provided camera poses, we skip SfM and train 3DGS on the augmented sets using default settings.

3 USER STUDY

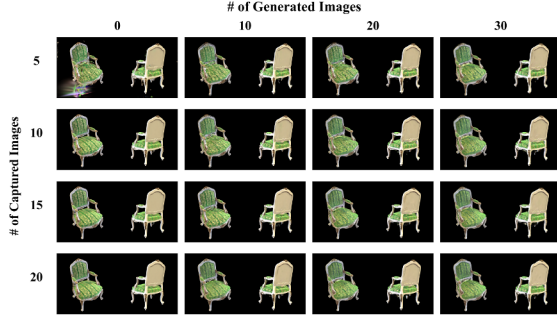


Figure 2: Example configurations for object *chair* with varying numbers of captured and generated images.

For each object, we constructed training sets using a small number of captured views (5, 10, 15, or 20) selected to cover the object from diverse directions. We then optionally augmented these sets by generating additional novel views (0, 10, 20, or 30) at camera poses that were not included in the captured subset. Figure 2 shows an example of the configurations. This resulted in 16 configurations per object and 64 configurations in total across the four objects.

Participants rated plausibility and visual quality on two 7-point Likert items using a web UI with four reference images and a fixed-trajectory slider (200 frames). The presentation order of configurations was randomized. Conditions were randomized in a repeated-measures study (N=20) with a brief practice phase.

4 RESULTS

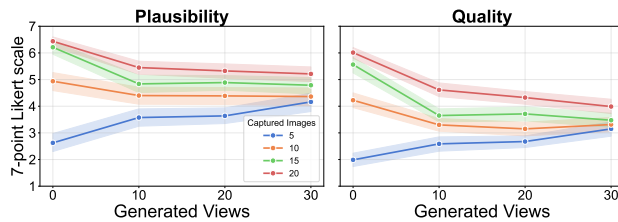


Figure 3: Mean ratings ($\pm 95\%$ CI) across objects for plausibility and quality versus the number of generated views.

Mean plausibility and visual quality ratings reveal a clear interaction between captured-view count and generative augmentation. When only 5-10 of captured images are available, adding generated views leads to noticeable improvements in both measures. As the number of captured views increases, however, this benefit diminishes, and for higher capture counts additional generated views can result in stable or even reduced ratings (Figure 3).

We fit a CLMM for ordinal repeated-measures ratings and report odds ratios vs. baseline, showing stronger effects in low-capture settings and weaker effects as capture coverage increases (Figure 4).

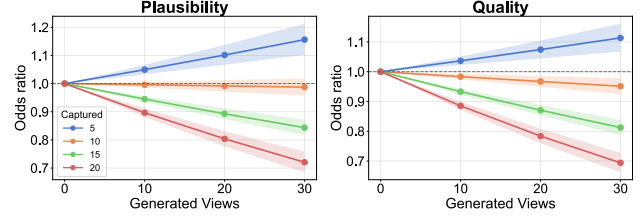


Figure 4: Average odds ratio by generated views across captured settings.

5 DISCUSSION AND CONCLUSION

Our results show that augmenting sparse training sets with generated views improves perceived plausibility and visual quality when captured images are 5-10. As capture coverage increases, gains diminish and can reverse when many generated views are added, consistent with both mean trends and CLMM. Practically, view generation helps few-shot capture, but should be used conservatively once real-view coverage becomes sufficient.

Several limitations point to clear directions for follow-up studies. The evaluation focuses on object-centric captures, a fixed viewing trajectory, and a single generation pipeline, and the participant count is modest. Future work should extend the analysis to a broader set of objects and capture conditions, include additional generators with stronger 3D awareness such as multi-view diffusion or world-model approaches, and test whether the thresholds generalize across generation methods. In addition to Likert ratings, paired comparisons can be incorporated to obtain more direct relative judgments between capture and augmentation settings and to strengthen ordering of perceived quality.

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